



United International University
School of Science and Engineering
Department of Computer Science and Engineering
Final Examination || Trimester: Spring – 2026
Course Title: Calculus and Linear Algebra (Math – 2183)
Marks – 40 || Duration – 2 hours

(The marks assigned to each question or part of a question are indicated in brackets []. Candidates are required to answer all questions and are advised to write their answers in the order in which the questions are presented)

Q1. (a) Solve $\begin{cases} 2x + y - z = 8 \\ x + 3y + 2z = 13 \\ 3x - y + z = 7 \end{cases}$ Using $X = A^{-1}B$. [5]

(c) For the system $\begin{cases} x + y + z = 1 \\ 2x + 3y + 4z = 2 \\ 3x + 4y + 5z = 3 \end{cases}$ [5]
find:
(i) Rank of coefficient matrix A
(ii) Rank of augmented matrix $[A | B]$
(iii) Determine consistency using the Rank Theorem.

Q2. (a) Solve the differential equation $(3x^2 + 10xy^2)dx + (10yx^2 + 3y^2)dy = 0$ [4]

(b) A damped mechanical system is subjected to an external force consisting of an exponentially decaying term and a periodic oscillatory term. If the damping coefficient is 6 and the restoring coefficient is 9, which form the differential equation. Find the displacement of the system.

$$y'' + 6y' + 9y = e^{-5x} + \sin 2x$$

Find the displacement $y(x)$ of the system. [6]

Q3. (a) A city traffic engineer is studying the movement of vehicles at a three-road intersection. The net flow of vehicles (in hundreds of vehicles per hour) along three different routes is modeled by the following vectors:

- Route A: $\mathbf{v}_1 = (1, 0, 2)$
- Route B: $\mathbf{v}_2 = (0, 1, 3)$
- Route C: $\mathbf{v}_3 = (2, 3, 13)$

Each vector represents traffic movement in three connected zones of the city.

- (i) Determine whether the three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent or dependent. [3]
- (ii) If they are linearly dependent, express one vector as a linear combination of the others. [2]

(b) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, enlarge points by scale factor -3 with center origin
(i) Find the standard matrix of the transformation. [1]
(ii) Triangle ABC has vertices $A(1, 2), B(3, 3), C(2, 4)$. Use the above transformation, find the image of the triangle ABC . [2]

Q4 (a) Given $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

(i) Find the eigenvalues of matrix A .

[2]

Verify that:

(ii) Sum of eigenvalues = trace of A

[1]

(iii) Product of eigenvalues = determinant of

[1]

(b) Find eigenvectors corresponding to each eigenvalue.

[4]

(c) Show that A is diagonalizable and find matrices P and D such that:

[3]

$$A = PDP^{-1}$$

(d) Use diagonalization to compute A^3 .

[1]